CVD DIAMOND DETECTORS OF IONIZING RADIATION

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Diamond possesses many attractive properties which make it an ideal material for particle detection in physics experiments as well as for medical dosimetry.

In 1981, CVD (chemical vapour deposition) diamond manufacturing technology was developed. This technology has the potential of allowing low cost production of diamond in large sheets and with higher purity than natural diamonds [1], [2]. Diamond has very high band gap, 5.5 eV, therefore it is an excellent insulator with high resistivity, more then 10^{11} (Ωcm). The high breakdown electric field of diamond (10^7 V/cm) allows application of high electric field and speeds up the charge collection. The electron and hole mobility is very high (higher than at GaAs and Si). The saturation velocity of diamond is high and allows high electron collection. The dielectric constant of diamond is low (5.7); it represents low detector capacitance at the input of the spectroscopic read-out amplifier. Therefore a diamond detector causes less noise than a geometrically identical Si detector [3]. Energy to form electron-hole pair is for diamond 13 eV. This is the only poor diamond property. The low atomic number of carbon can be an advantage in two ways. The first: reducing high-energy cascades and multiple scattering in physics experiments. The second: absorption characteristics are well matched to human soft tissue, so that diamond (Z=6) can be considered soft-tissue-equivalent (Z=7.42) [4]. High radiation hardness (1MGy) makes diamond an ideal material, especially in the high radiation environments of future colliders like the LHC in CERN.

CVD diamond coaxial X-ray microdetectors are prepared by CVD technology. CVD diamond film is deposited by a hot filament CVD technique on metallic(W, Mo) wires and tips. The diamond film (10 µm thick) covers uniformly the W (tungsten) tip and is used as an active semiconductor region for X-ray detection. A graphite layer covers the diamond film and W substrate wire acts as an electrode. Coaxial detectors are stable and relatively fast. Their signal-to-noise ratio is approximately 1000 [5]. Coaxial detectors can be used for monitoring X-rays and γ-rays in medical and health physics fields. Coaxial geometry is more suitable for probe miniaturization. Probes can have diameter up to 50 µm.

CVD diamond micro-strip detectors have a pattern of strips with 50µm pitch on the irradiated side of diamond and a guide ring around this structure. The other surface of diamond is covered by a uniform metallic contact. For contacts, chromium is used as a rule, covered by a gold layer to produce the ohmic contact [3]. Strip detectors are used to track particle paths. An incident particle is recorded by a strip which generates the signal. Strip detectors were used in CERN to study the mechanism of charge collection and to measure the spatial resolution of diamond detectors by a pion beam. The first diamond strip detectors with a 2*4 cm² area have been produced. These can be used in real physics experiments. The signal-to-noise ratio was 23/1 and the spatial resolution was 14 µm.

The first 16*16 matrix of 150*150 µm² diamond pixel detector was tested in 1996 [3]. The pixels were wire-bonded to a fan-out on glass substrate and read-out using VA3 chip. The measured signal-to-noise ratio was 27/1 and the position resolution was consistent with the digital resolution of the pixels. The development of diamond pixel detectors has made good progress in the difficult area of detector metallization for connection with the bump-bonding technique. The first results from a pixel detector (ATLAS3) in a test beam demonstrated 98% efficiency for the bump-bonding of the pixels and spatial resolution for the track measurement in both dimension.

Recently, diamond layer technology has been successfully applied at the Slovak University of Technology [6]. In collaboration with our DNPT and Department of Microelectronics, diamond detectors for dosimetry applications are currently being prepared.

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